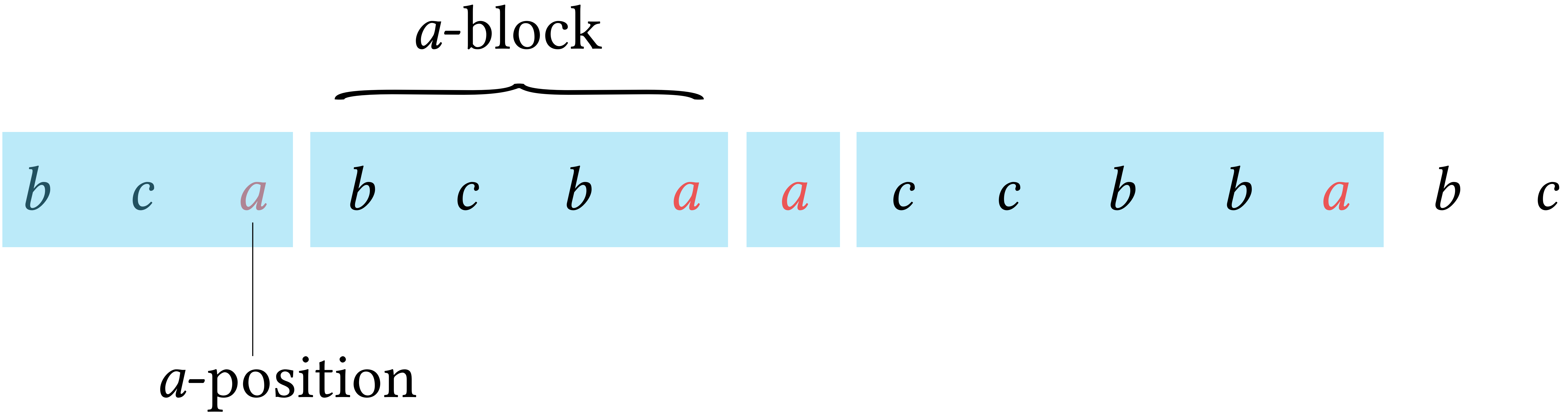
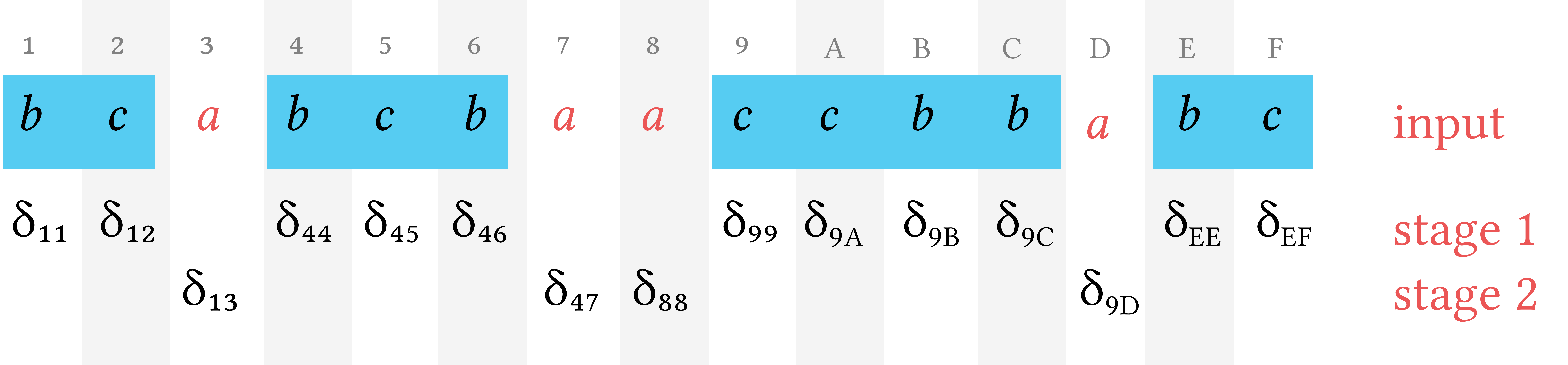

a-block
b c a
b c b a
a
c c b b a
b c
a-position

$$\underbrace{b\ c\ a}_{\substack{\text{first part:}\\ \text{first } a\text{-block}}}\ \underbrace{b\ c\ b\ a\quad a\quad c\ c\ b\ b\ a}_{\substack{\text{middle part:}\\ \text{all } a\text{-blocks except the first}}}\ \underbrace{b\ c}_{\substack{\text{last part:}\\ a\text{-free suffix}}}$$

	1	2	3	4	5	6	7	8	9	A	B	C	D	E	F
input	b	c	a	b	c	b	a	a	c	c	b	b	a	b	c
stage 1	δ_{11}	δ_{12}		δ_{44}	δ_{45}	δ_{46}			δ_{99}	δ_{9A}	δ_{9B}	δ_{9C}		δ_{EE}	δ_{EF}

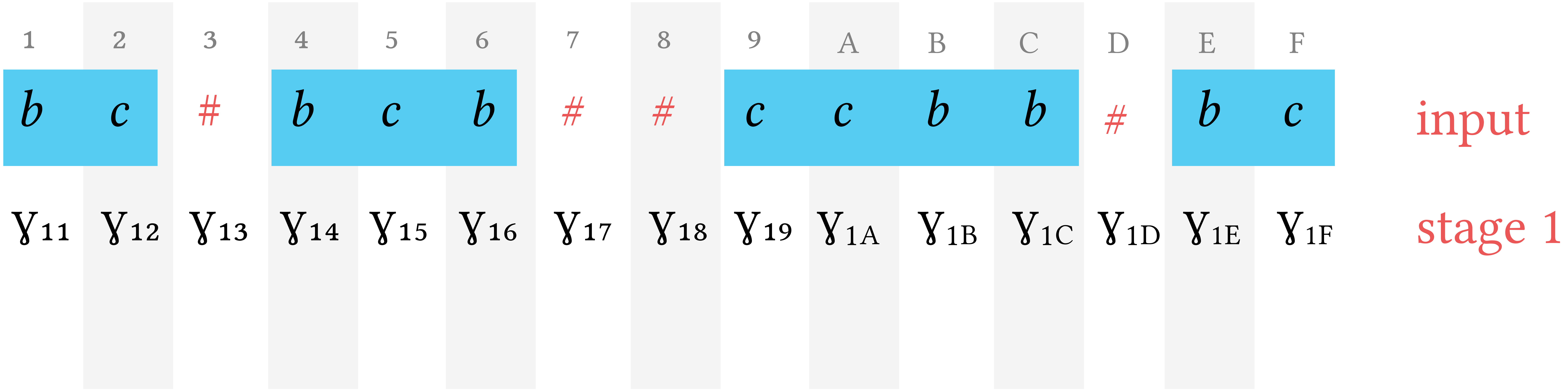


1 2 3 4 5 6 7 8 9 A B C D E F
b c a b c b a a c c b b a b c
input
δ11 δ12 δ44 δ45 δ46 δ99 δ9A δ9B δ9C δEE δEF
stage 1
δ13 δ47 δ88 δ9D
stage 2

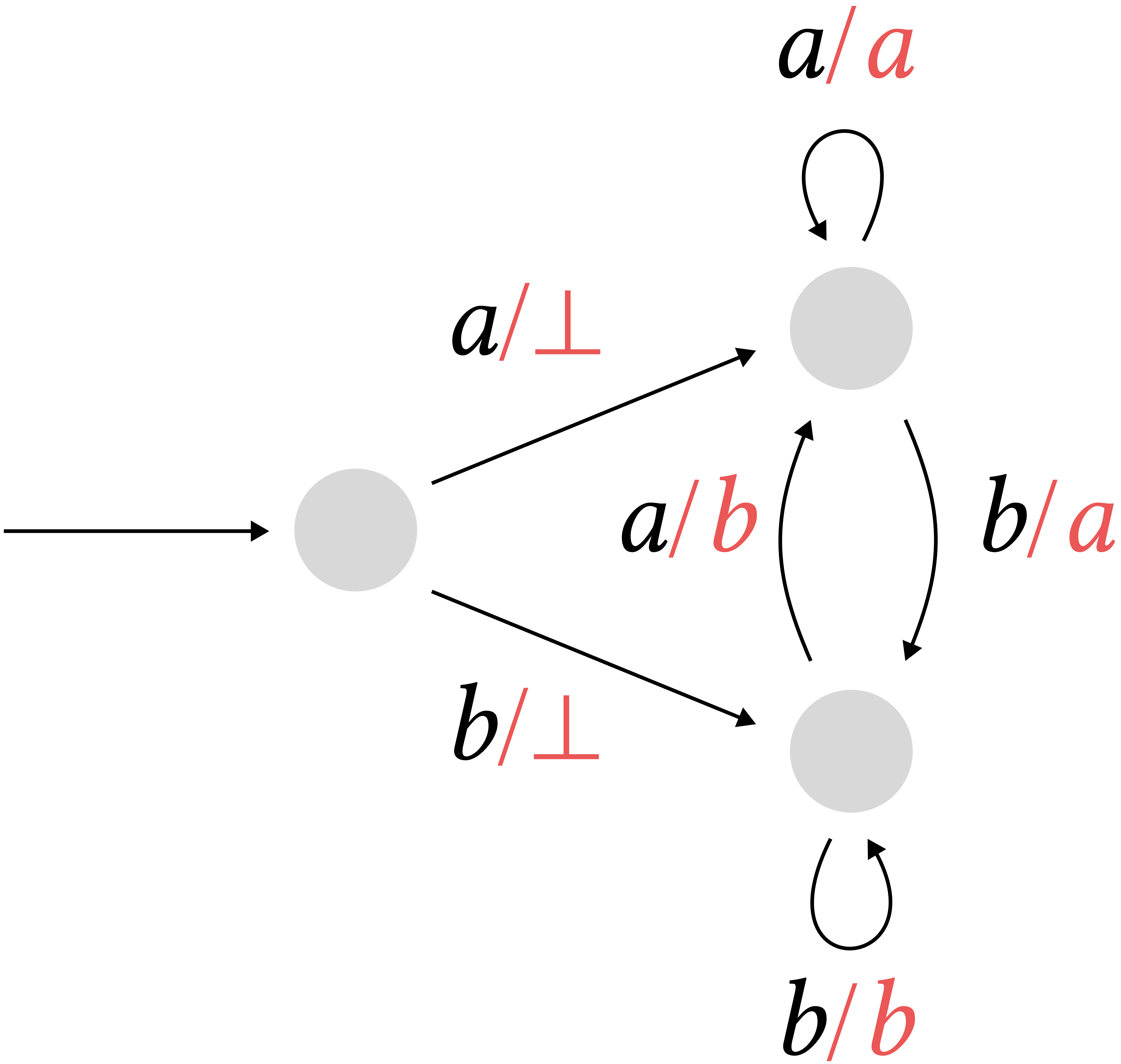
	1	2	3	4	5	6	7	8	9	A	B	C	D	E	F
input	b	c	a	b	c	b	a	a	c	c	b	b	a	b	c
stage 1	δ_{11}	δ_{12}		δ_{44}	δ_{45}	δ_{46}			δ_{99}	δ_{9A}	δ_{9B}	δ_{9C}		δ_{EE}	δ_{EF}
stage 2			δ_{13}				δ_{37}	δ_{88}					δ_{9D}		
stage 3				δ_{13}	δ_{13}	δ_{13}	δ_{13}	δ_{13}	δ_{13}	δ_{13}	δ_{13}	δ_{13}	δ_{13}	δ_{13}	δ_{13}

b	c	a	b	c	b	a	a	c	b	input
$\perp$	b	c	a	b	c	b	a	a	c	output

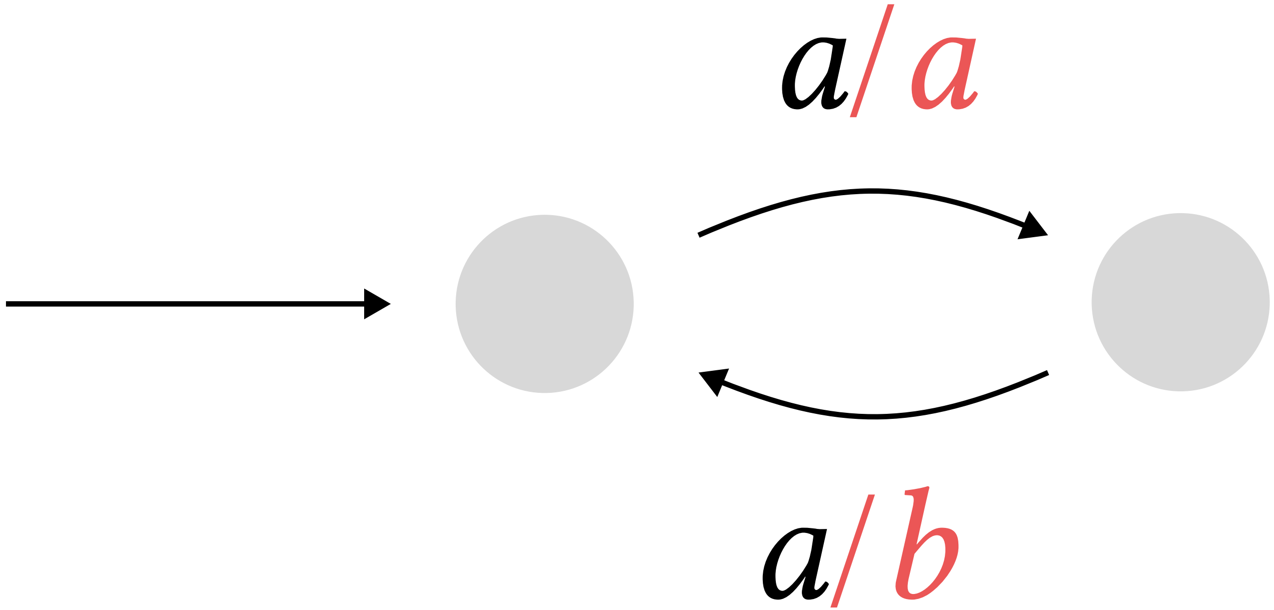
a	a	a	a	a	a	a	a	a	a	input
a	b	a	b	a	b	a	b	a	b	output


1 2 3 4 5 6 7 8 9 A B C D E F
b c # b c b # # c c b b # b c
input
γ11 γ12 γ13 γ14 γ15 γ16 γ17 γ18 γ19 γ1A γ1B γ1C γ1D γ1E γ1F
stage 1

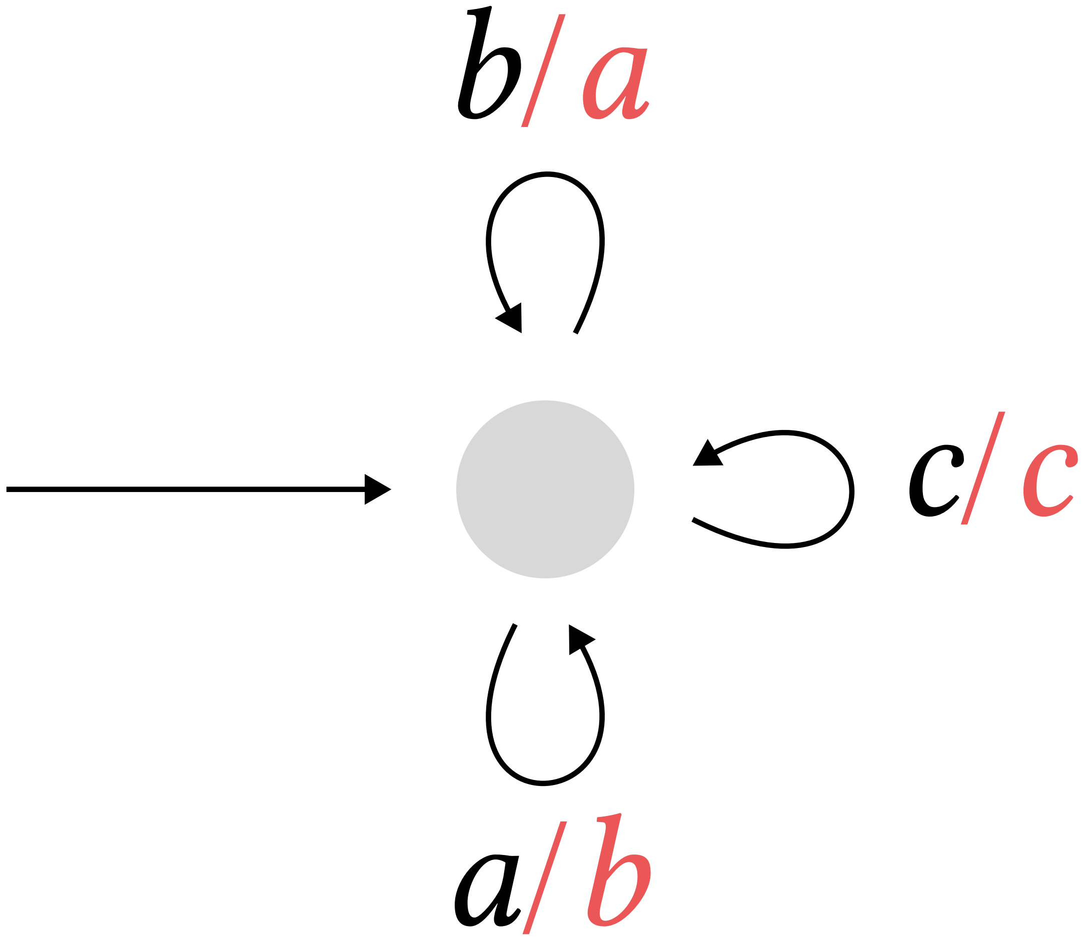
	1	2	3	4	5	6	7	8	9	A	B	C	D	E	F
input	b	c	#	b	c	b	#	#	c	c	b	b	#	b	c
stage 1	γ_{11}	γ_{12}	γ_{13}	γ_{14}	γ_{15}	γ_{16}	γ_{17}	γ_{18}	γ_{19}	γ_{1A}	γ_{1B}	γ_{1C}	γ_{1D}	γ_{1E}	γ_{1F}
stage 2				γ_{13}	γ_{13}	γ_{13}	γ_{13}	γ_{17}	γ_{18}	γ_{18}	γ_{18}	γ_{18}	γ_{18}	γ_{1D}	γ_{1D}



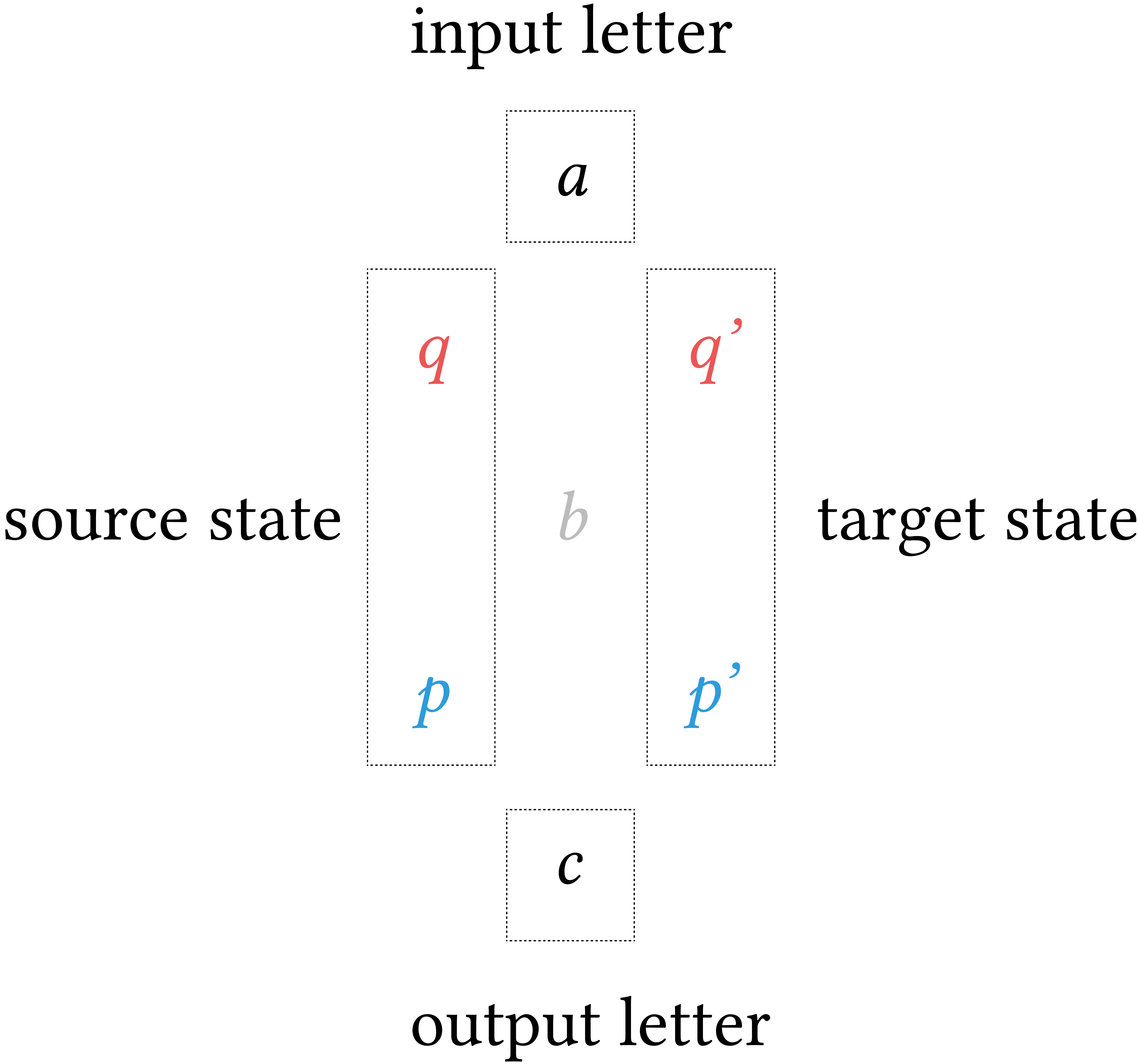
a/a
a/⊥
a/b
b/a
b/⊥
b/b



a/a
a/b


b/a
c/c
a/b

$$
\begin{array}{cccccccccccccccccccc}
 & a_1 & & a_2 & & a_3 & & a_4 & & a_5 & & a_6 & & a_7 & & a_8 & & a_9 & & \\
q_0 & & q_1 & & q_2 & & q_3 & & q_4 & & q_5 & & q_6 & & q_7 & & q_8 & & q_9 & \text{run of } f \\
 & b_1 & & b_2 & & b_3 & & b_4 & & b_5 & & b_6 & & b_7 & & b_8 & & b_9 & & \\
p_0 & & p_1 & & p_2 & & p_3 & & p_4 & & p_5 & & p_6 & & p_7 & & p_8 & & p_9 & \text{run of } g \\
 & c_1 & & c_2 & & c_3 & & c_4 & & c_5 & & c_6 & & c_7 & & c_8 & & c_9 & &
\end{array}
$$


input letter
a
q
q'
source state
b
target state
p
p'
c
output letter

b	c	a	b	c	b	a	a	c	b	input
$\perp$	$\perp$	b	c	a	b	c	b	a	a	output

$b \quad c \quad a \quad b \quad c \quad b \quad a \quad a \quad c \quad b$

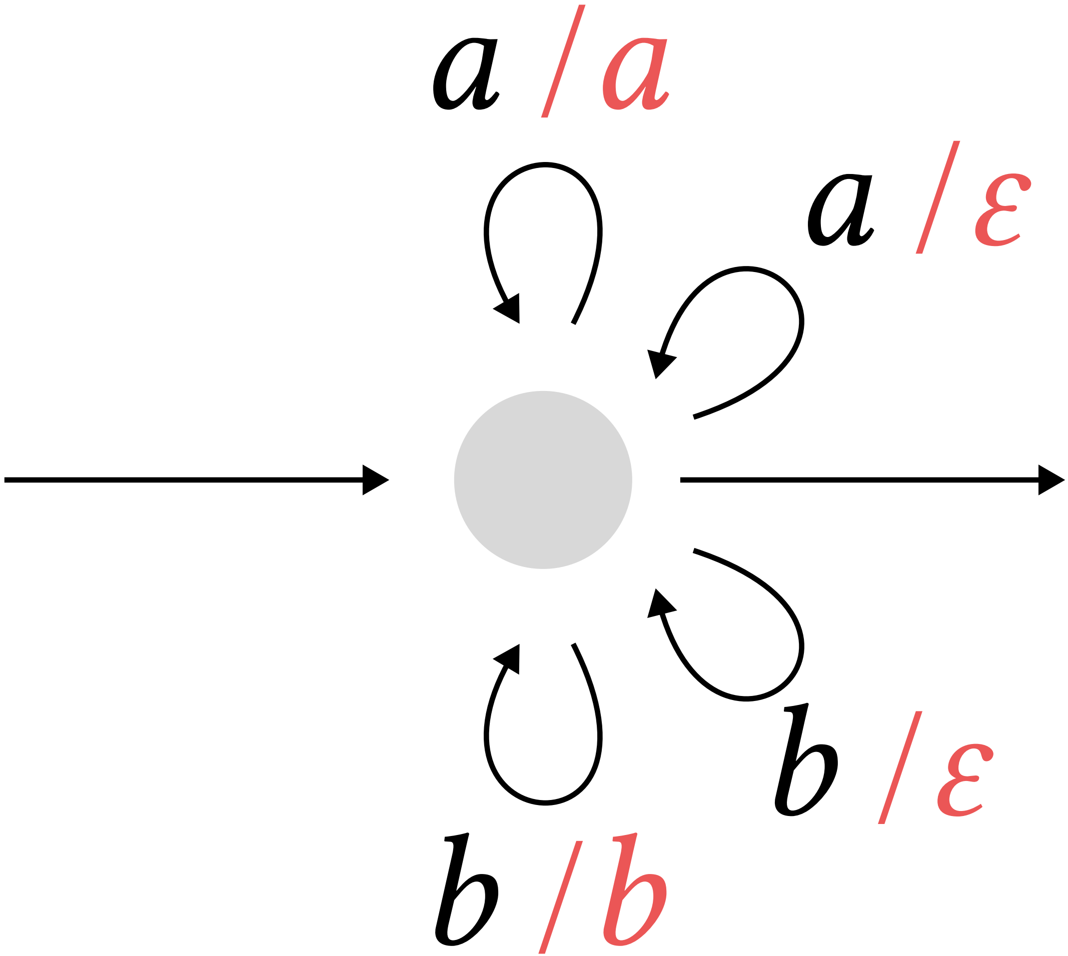
$\longrightarrow p \quad q \longleftarrow$

a *prefix* automaton is run from left-to-right

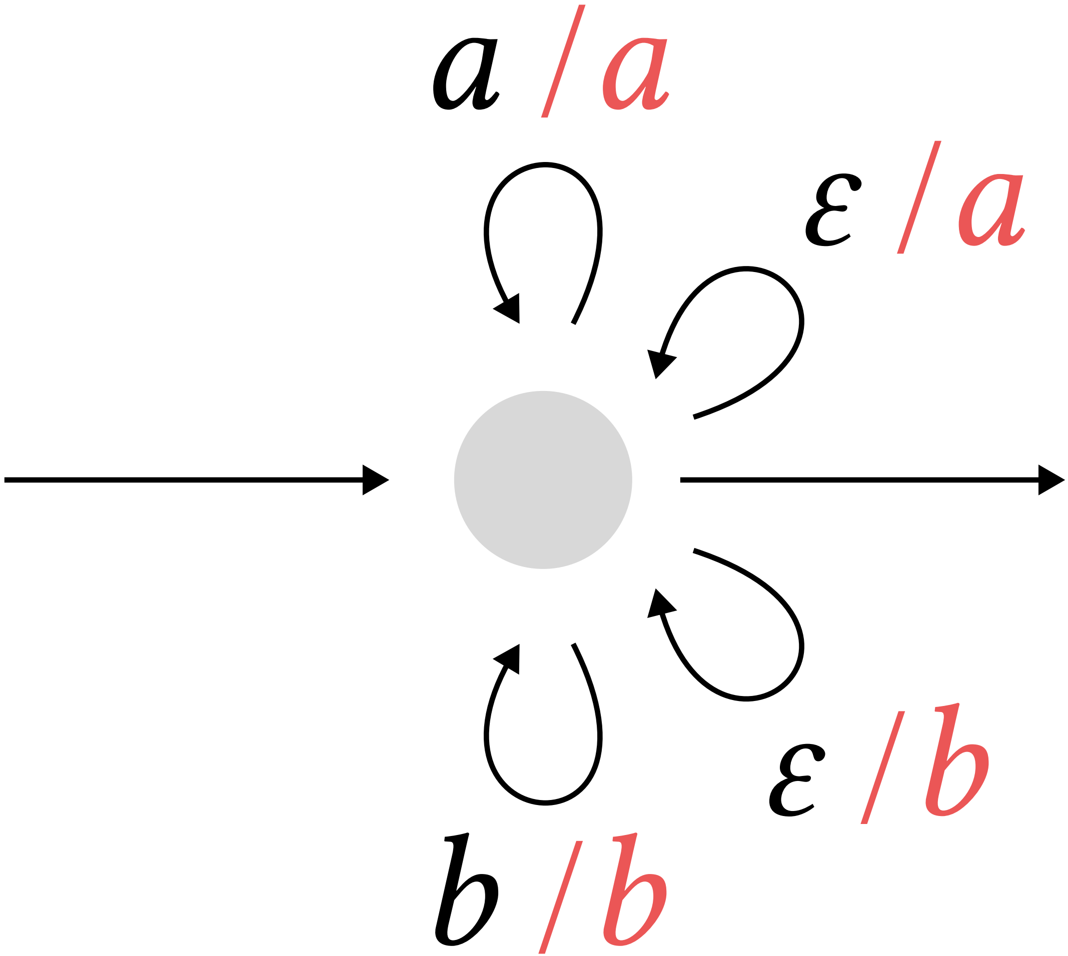
a *suffix* automaton is run from right-to-left

$$cdd$$

based on the states of the prefix and suffix automata, a piece of the output is produced

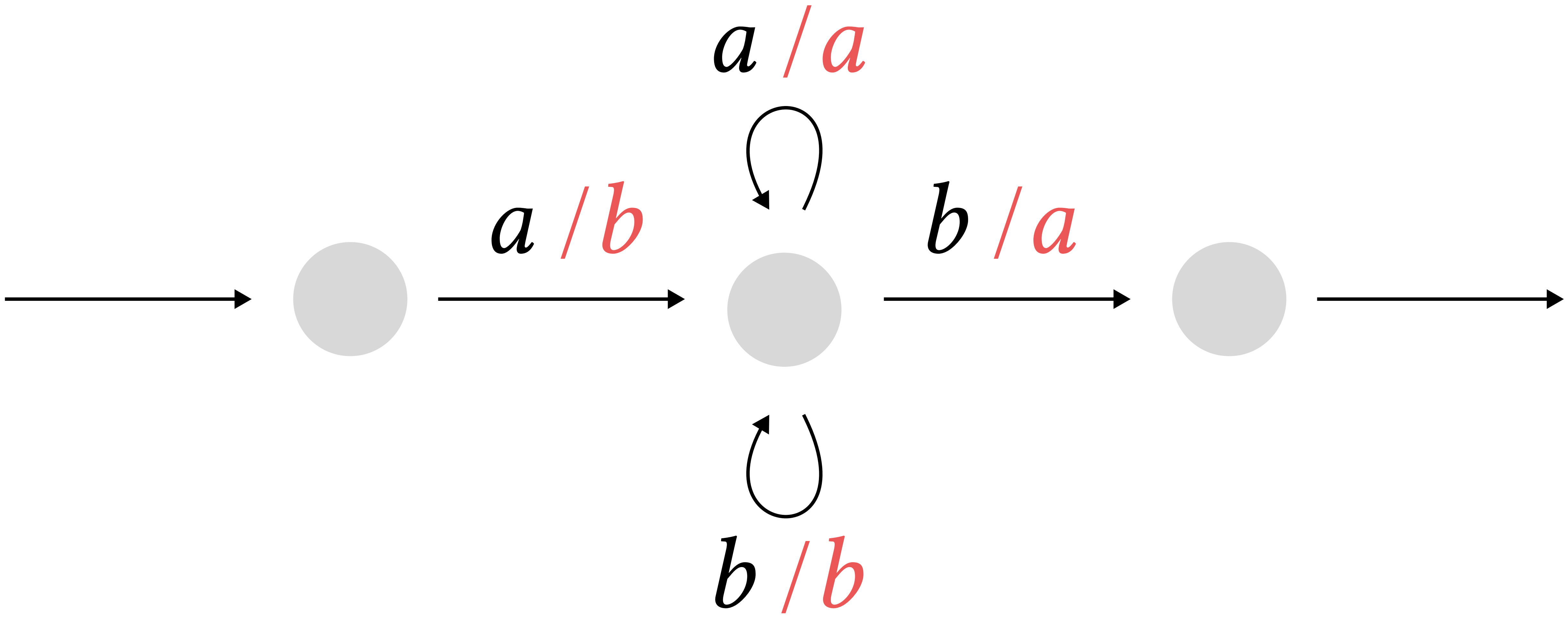


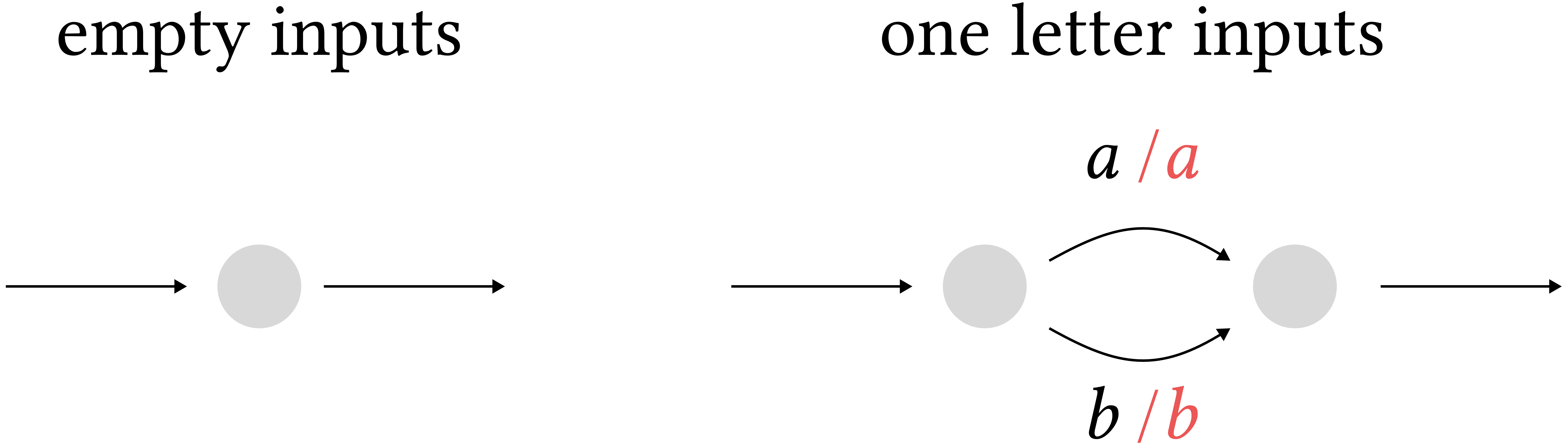
a / a
a / ε
b / ε
b / b

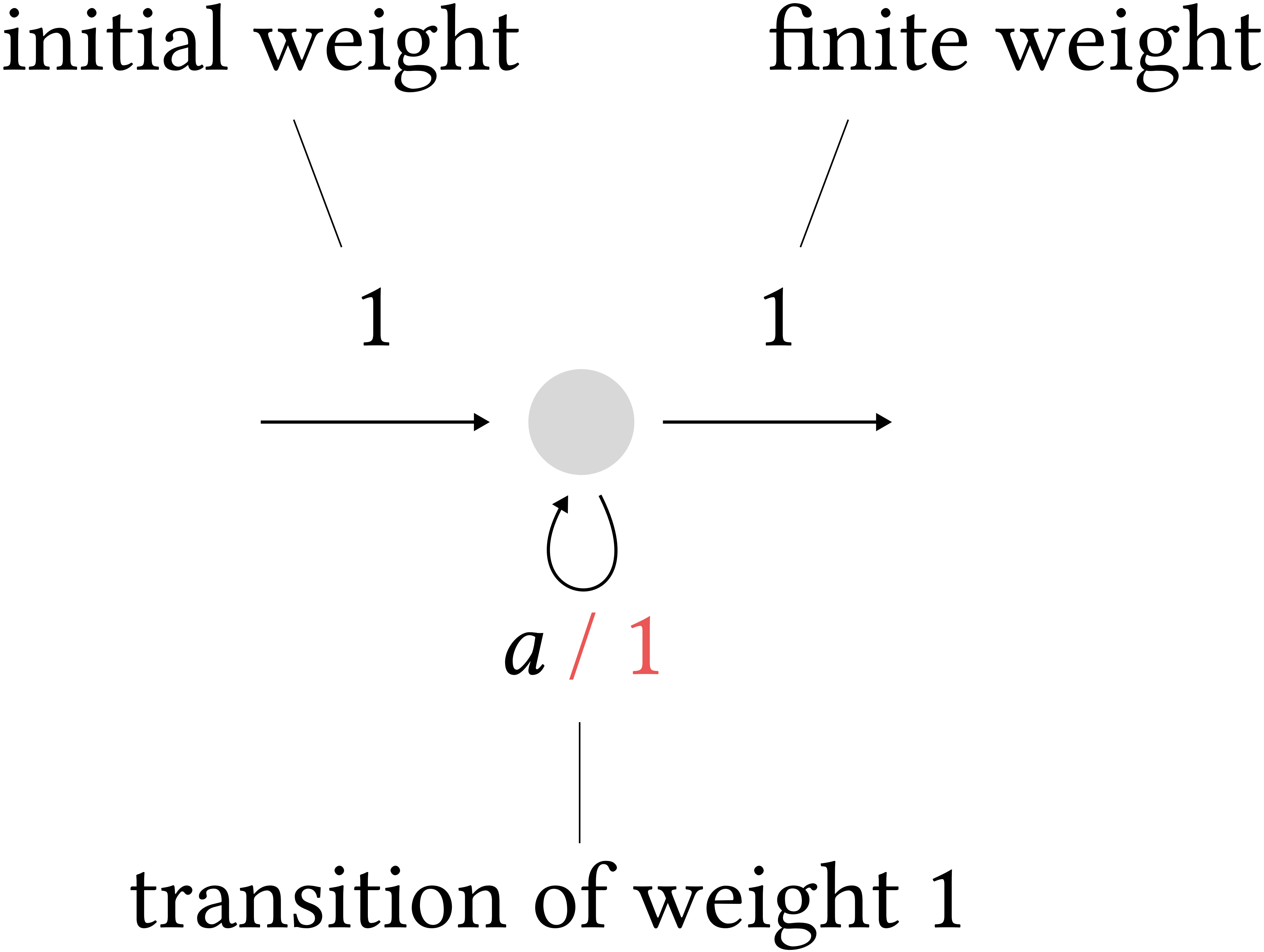


a / a
ε / a
ε / b
b / b

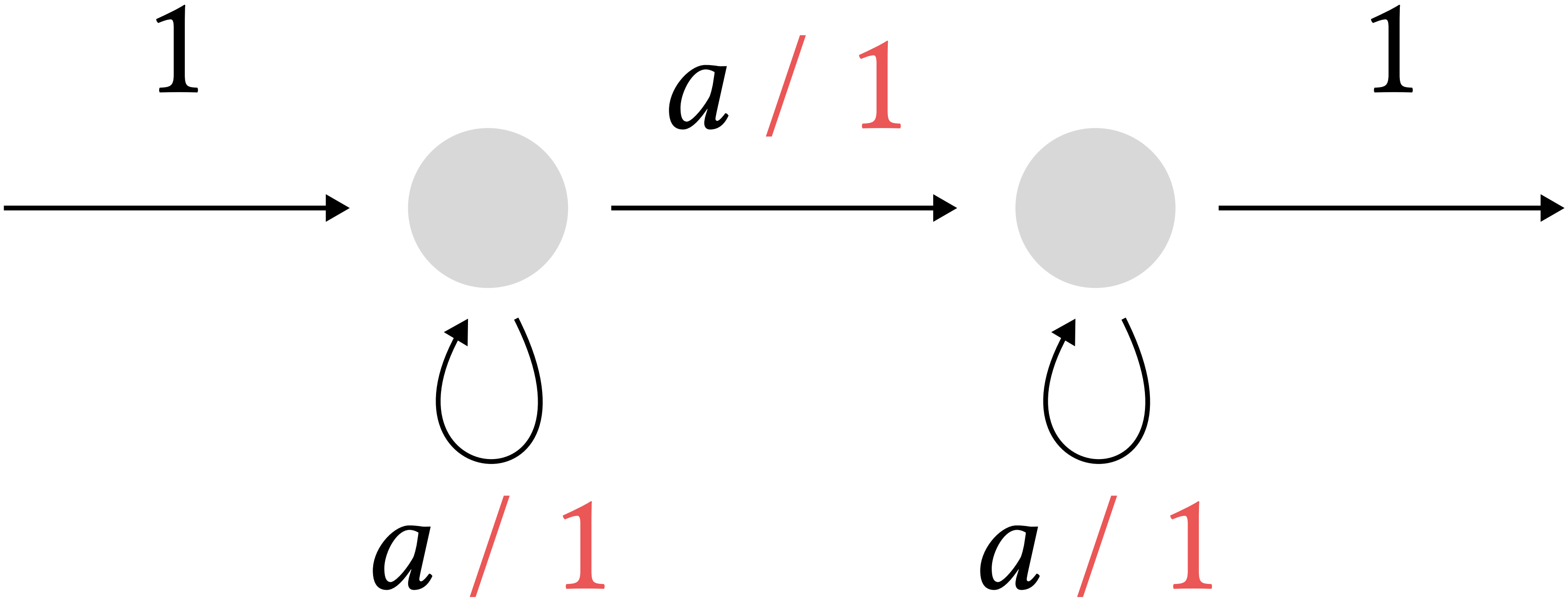
inputs that start with f and end with b

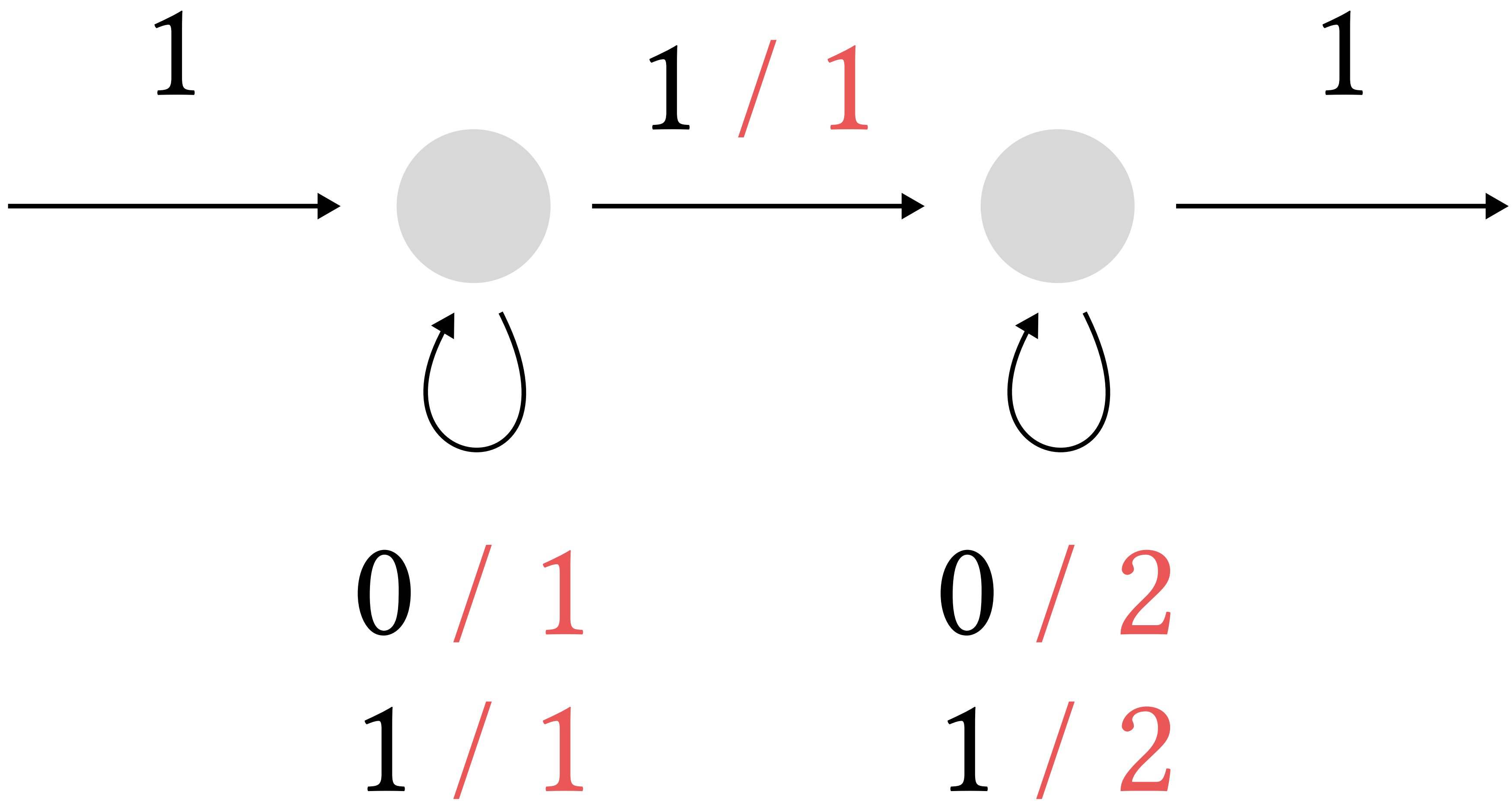



empty inputs
one letter inputs
a / a
b / b

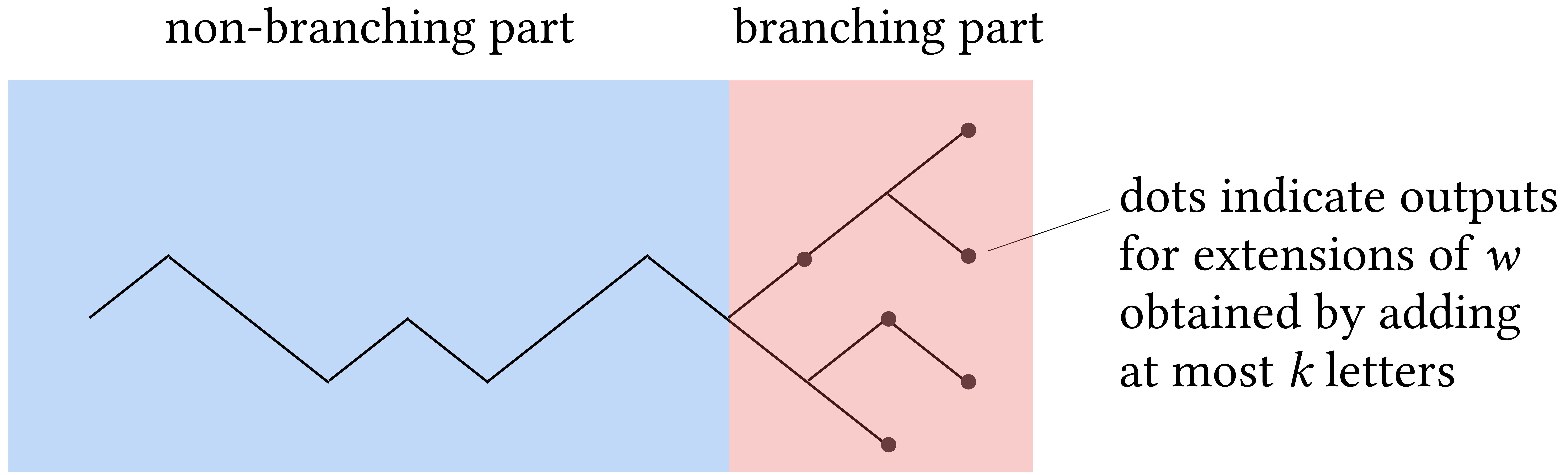


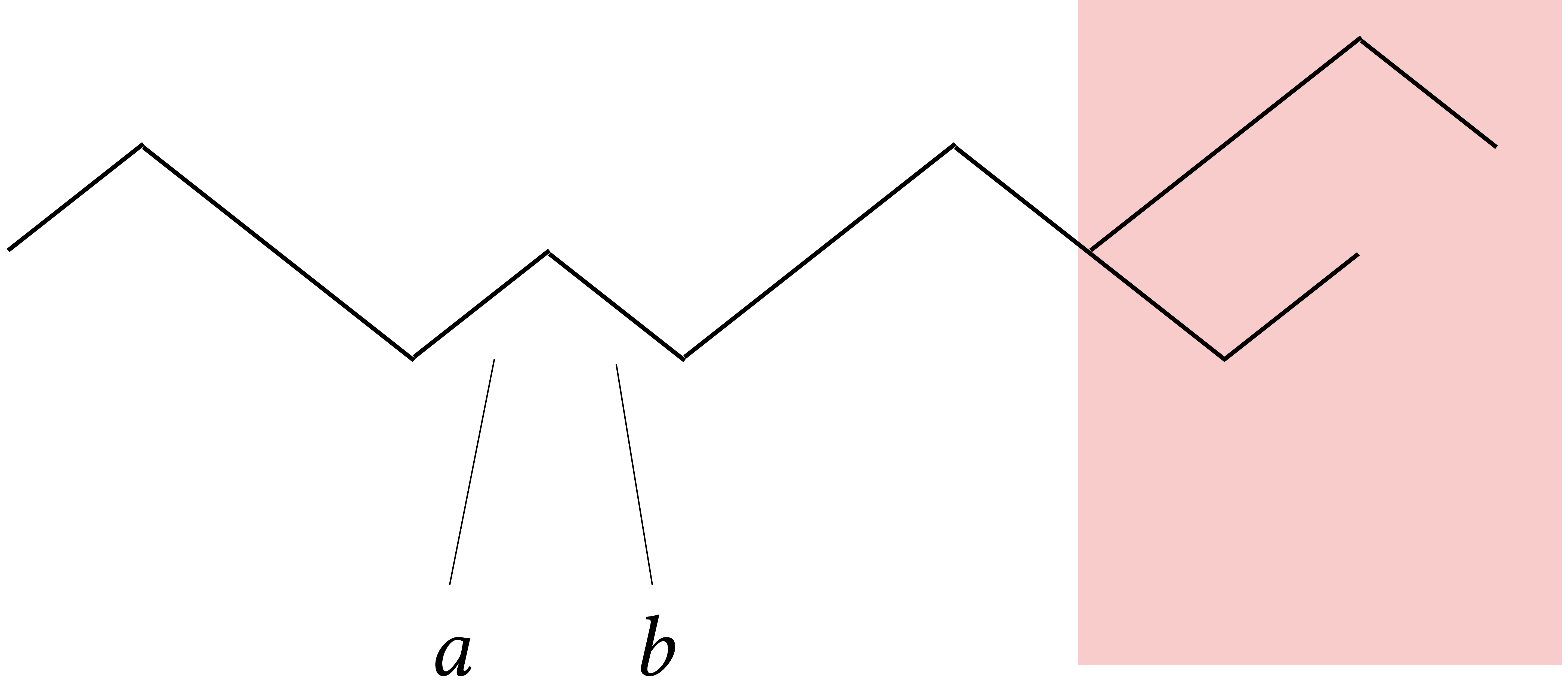
initial weight
finite weight
1
1
a / 1
transition of weight 1


1
a / 1
1
a / 1
a / 1


1
1 / 1
1
0 / 1
1 / 1
0 / 2
1 / 2

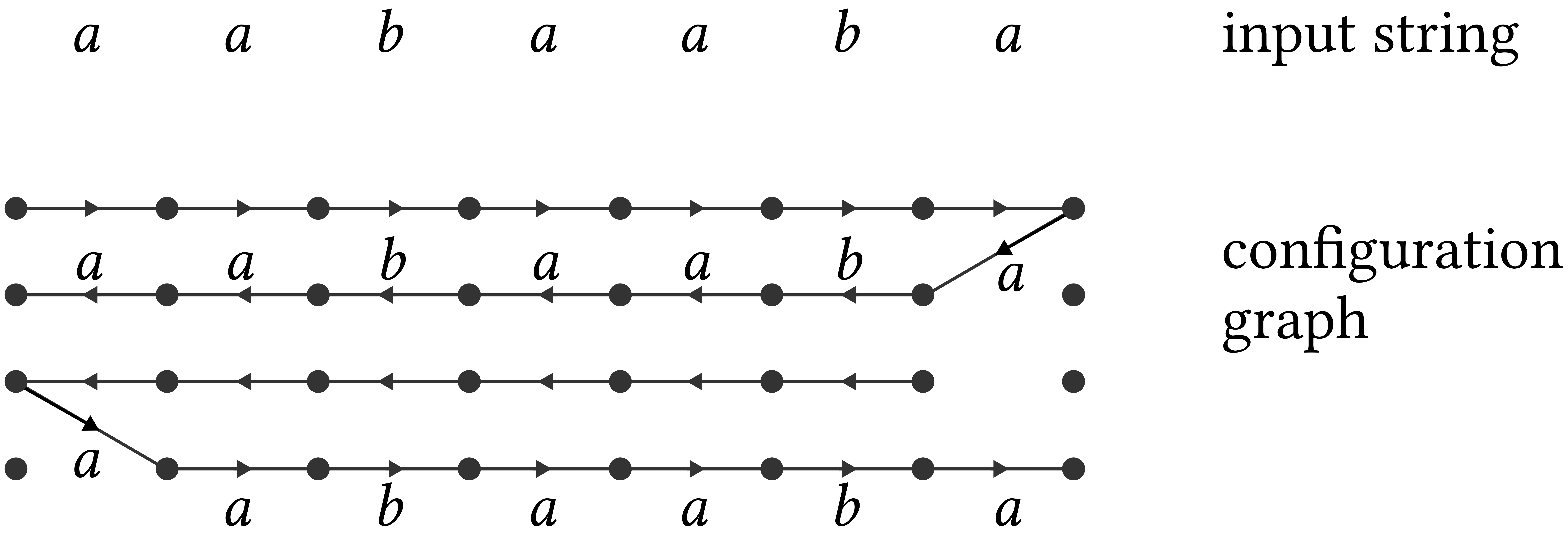
$$p \xrightarrow{\varepsilon \,/\, cd} q \xrightarrow{\varepsilon \,/\, \varepsilon} r \xrightarrow{a \,/\, c} r \xrightarrow{\varepsilon \,/\, cd} p \xrightarrow{b \,/\, c} r \xrightarrow{a \,/\, cd} q \xrightarrow{\varepsilon \,/\, c} r \xrightarrow{\varepsilon \,/\, c} p$$


non-branching part
branching part
dots indicate outputs
for extensions of w
obtained by adding
at most k letters



directions indicate letters left distance $k = 3$

a a b # b a b # b b
input string
a a b
b a b
b b
#
#
a
a b
b
a b
b
b
configuration graph



a a b a a b a
input string
a a b a a b a
configuration
graph
a
a
a b a a b a

$$a \quad b \quad a \quad a \quad a \quad b \quad a \quad a$$

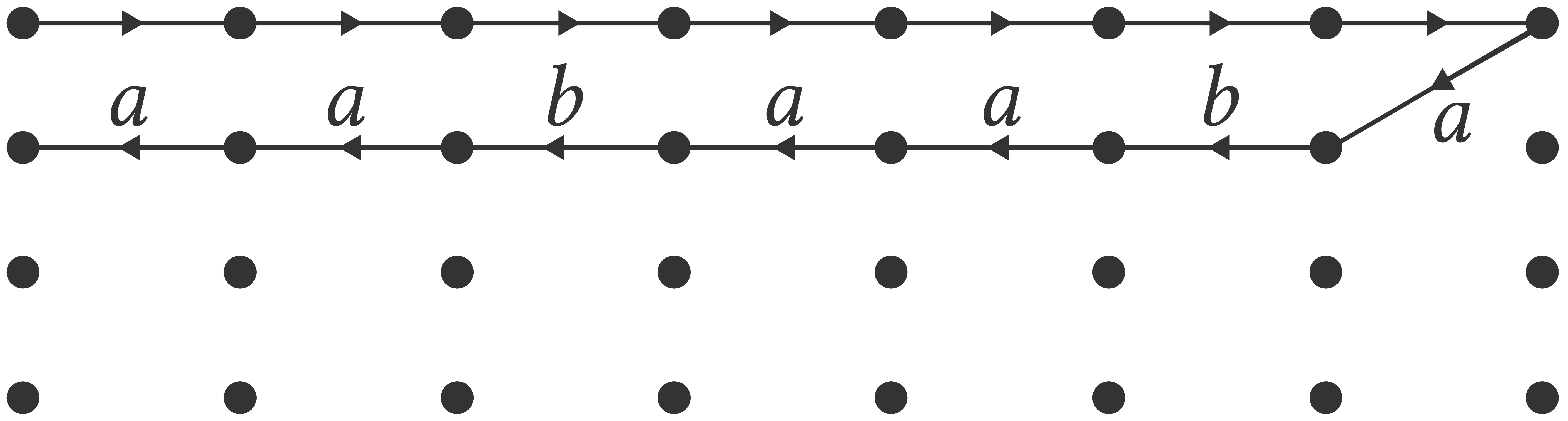
$$\uparrow$$

$$q$$

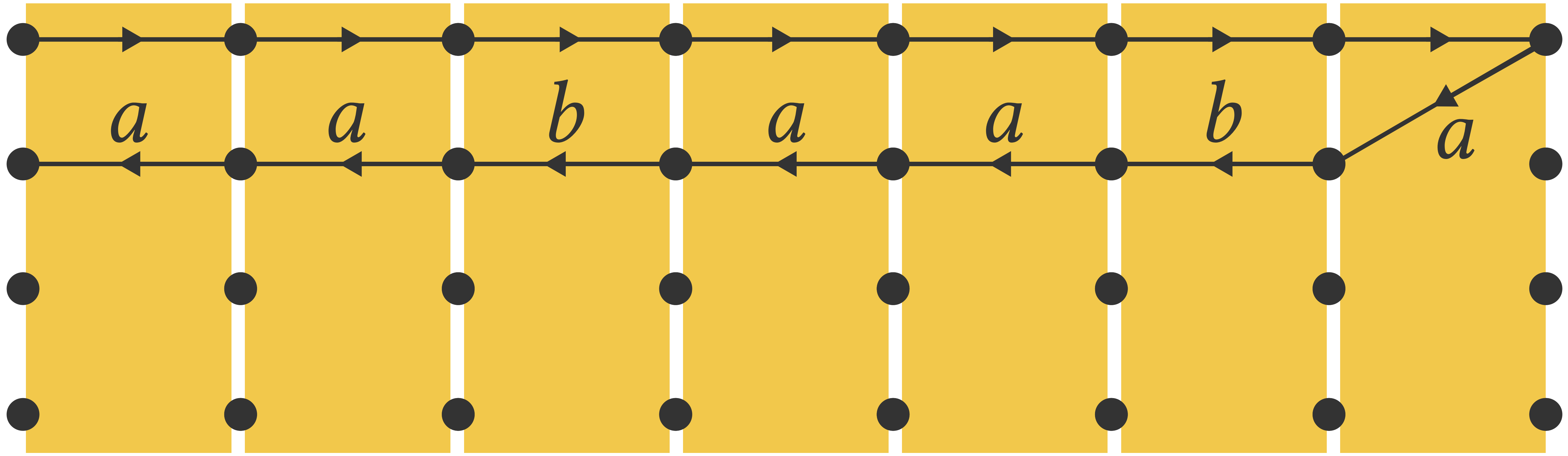
$a \quad b \quad a \quad a \quad a \quad b \quad a \quad a$

q : state of Mealy machine f

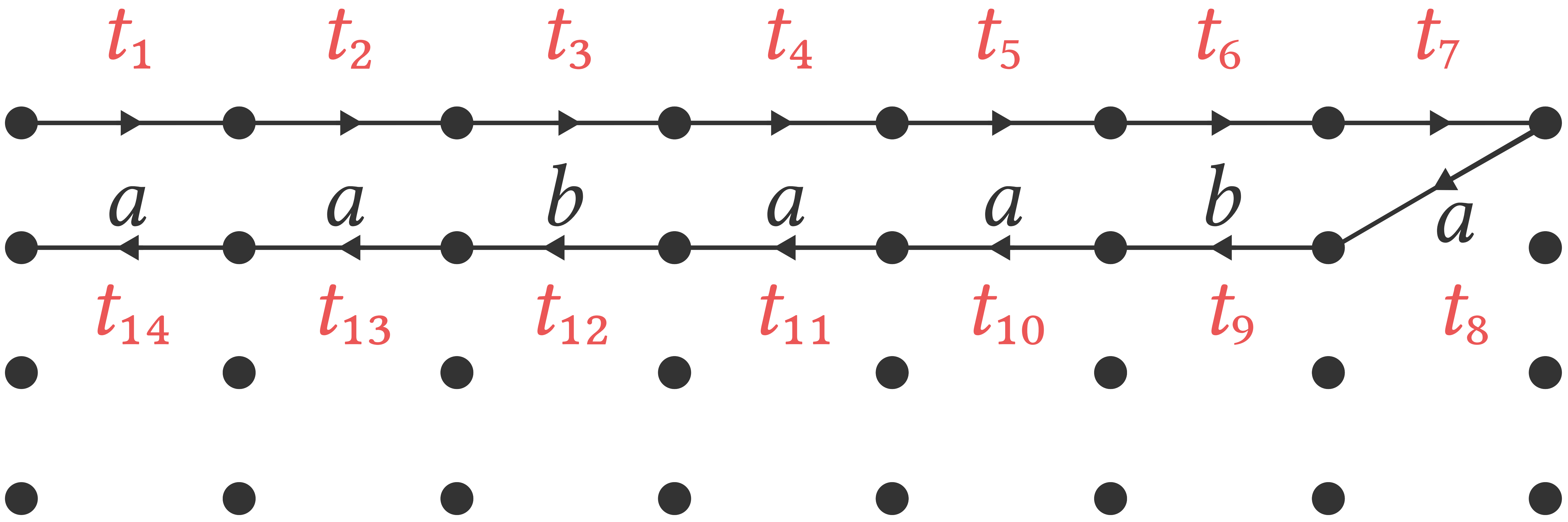
p : state of two-way transducer g



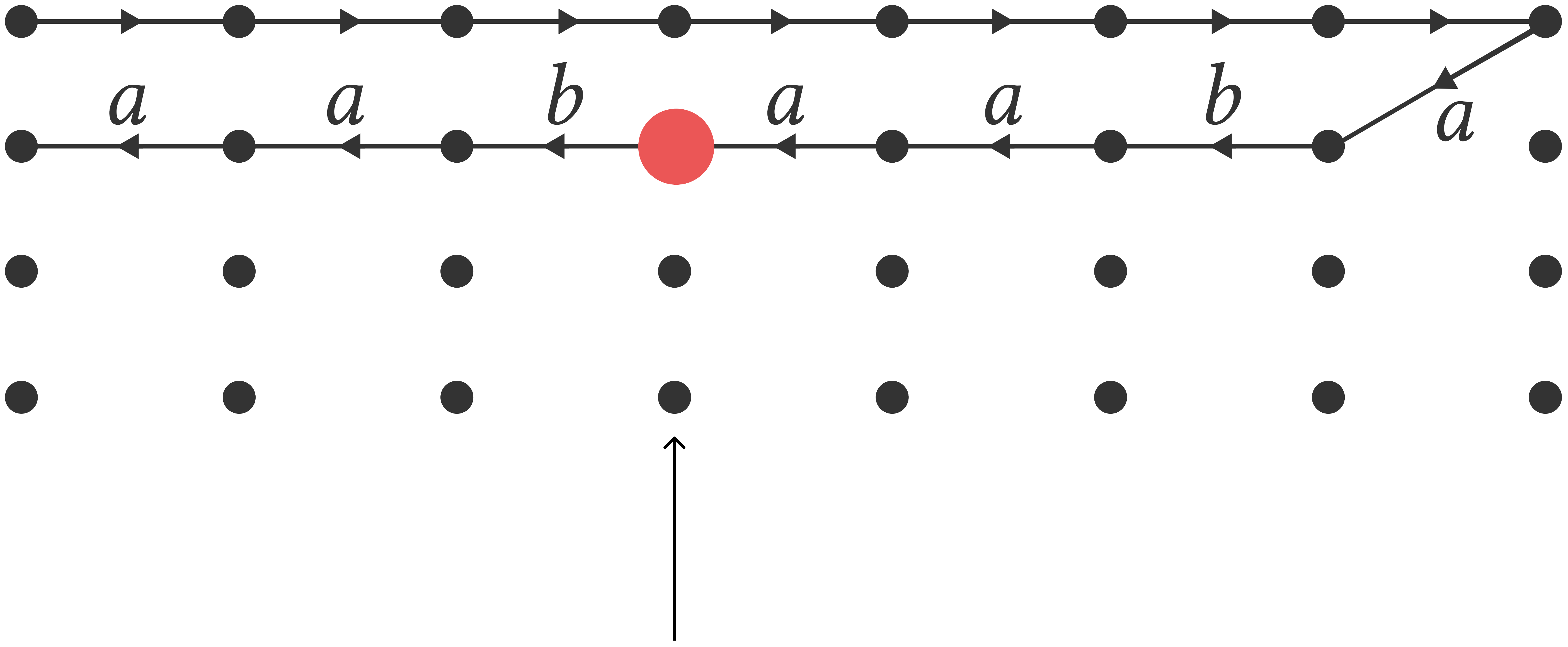
reachable part of the configuration graph



string representation



t_1
t_2
t_3
t_4
t_5
t_6
t_7
a
a
b
a
a
b
a
t_{14}
t_{13}
t_{12}
t_{11}
t_{10}
t_9
t_8



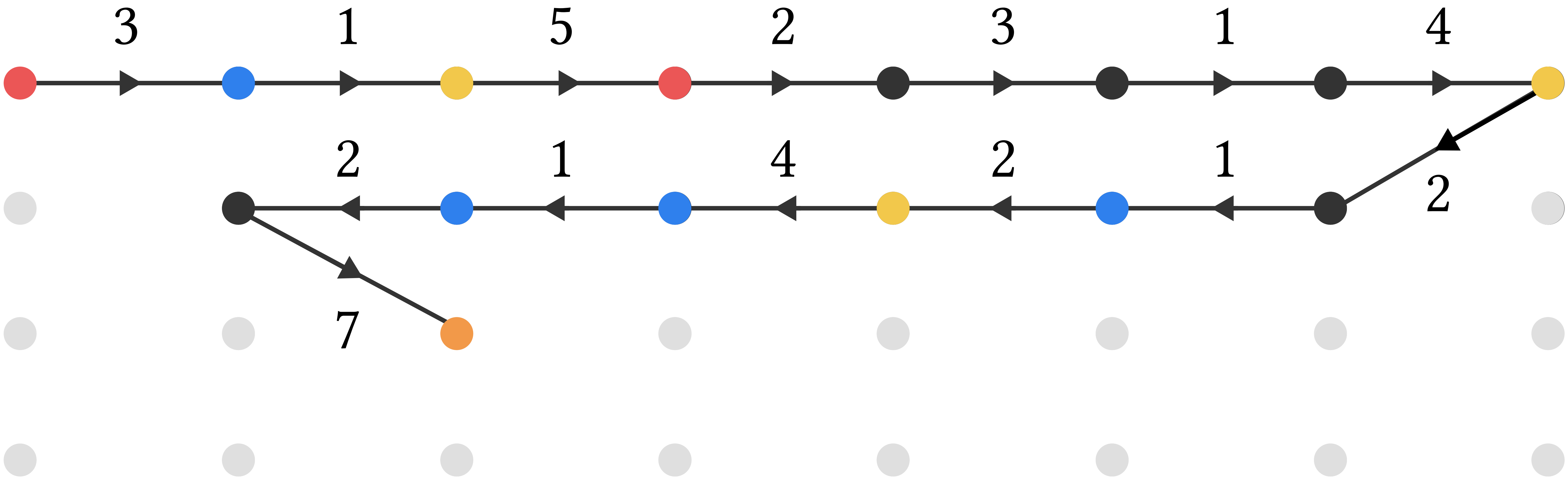
product automaton remembers:

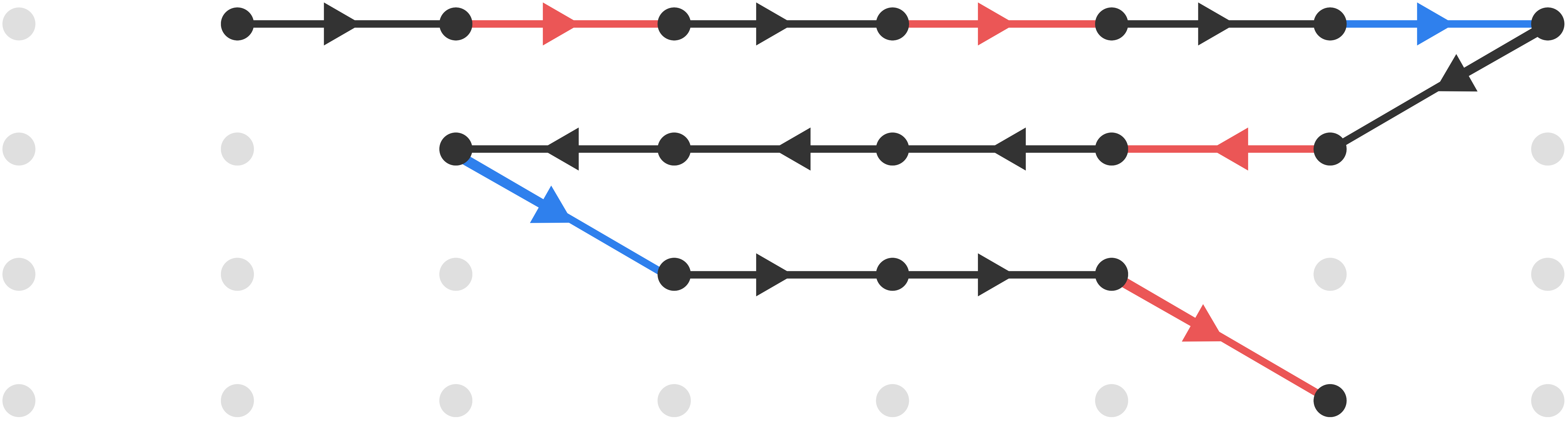
- state of f (red circle); and
- state of g

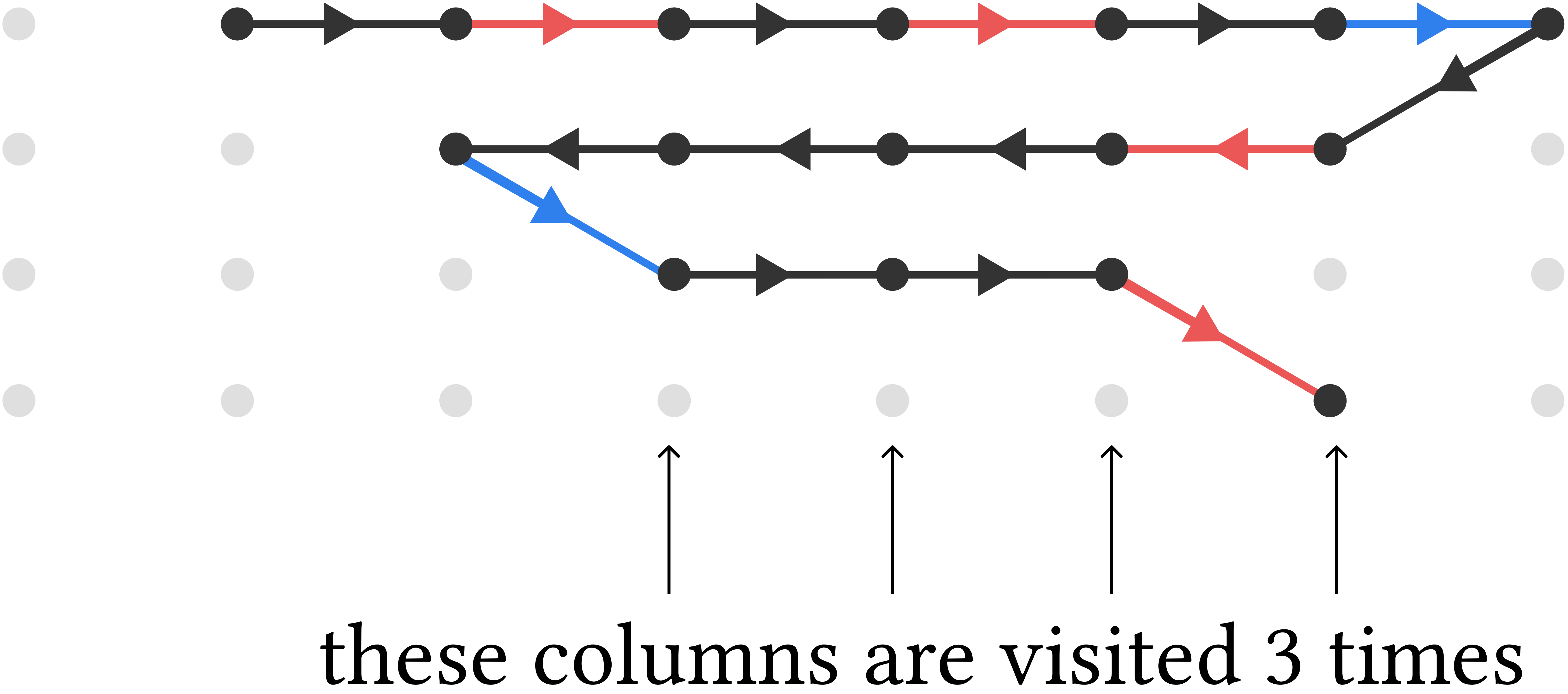
$$
\begin{tikzcd}
q_1 \arrow[r, phantom, "="] \arrow[d, equal] & q_1 \arrow[drr] & q_5 \arrow[r, phantom, "="] \arrow[d, equal] & q_5 \arrow[r, phantom, "="] & q_5 \arrow[r, phantom, "="] & q_5 \arrow[drr] & q_9 \arrow[r, phantom, "="] \arrow[d, equal] & q_9 \arrow[r, phantom, "="] & q_9 \arrow[r, phantom, "="] & q_9 \\
q_1 & q_0 \arrow[l] \arrow[d, equal] & q_5 & q_4 \arrow[l] & q_3 \arrow[l] & q_2 \arrow[l] \arrow[d, equal] & q_9 & q_8 \arrow[l] & q_7 \arrow[l] & q_6 \arrow[l] \arrow[d, equal] \\
q_0 \arrow[r, phantom, "="] & q_0 & q_2 \arrow[r, phantom, "="] & q_2 \arrow[r, phantom, "="] & q_2 \arrow[r, phantom, "="] & q_2 & q_6 \arrow[r, phantom, "="] & q_6 \arrow[r, phantom, "="] & q_6 \arrow[r, phantom, "="] & q_6
\end{tikzcd}
$$

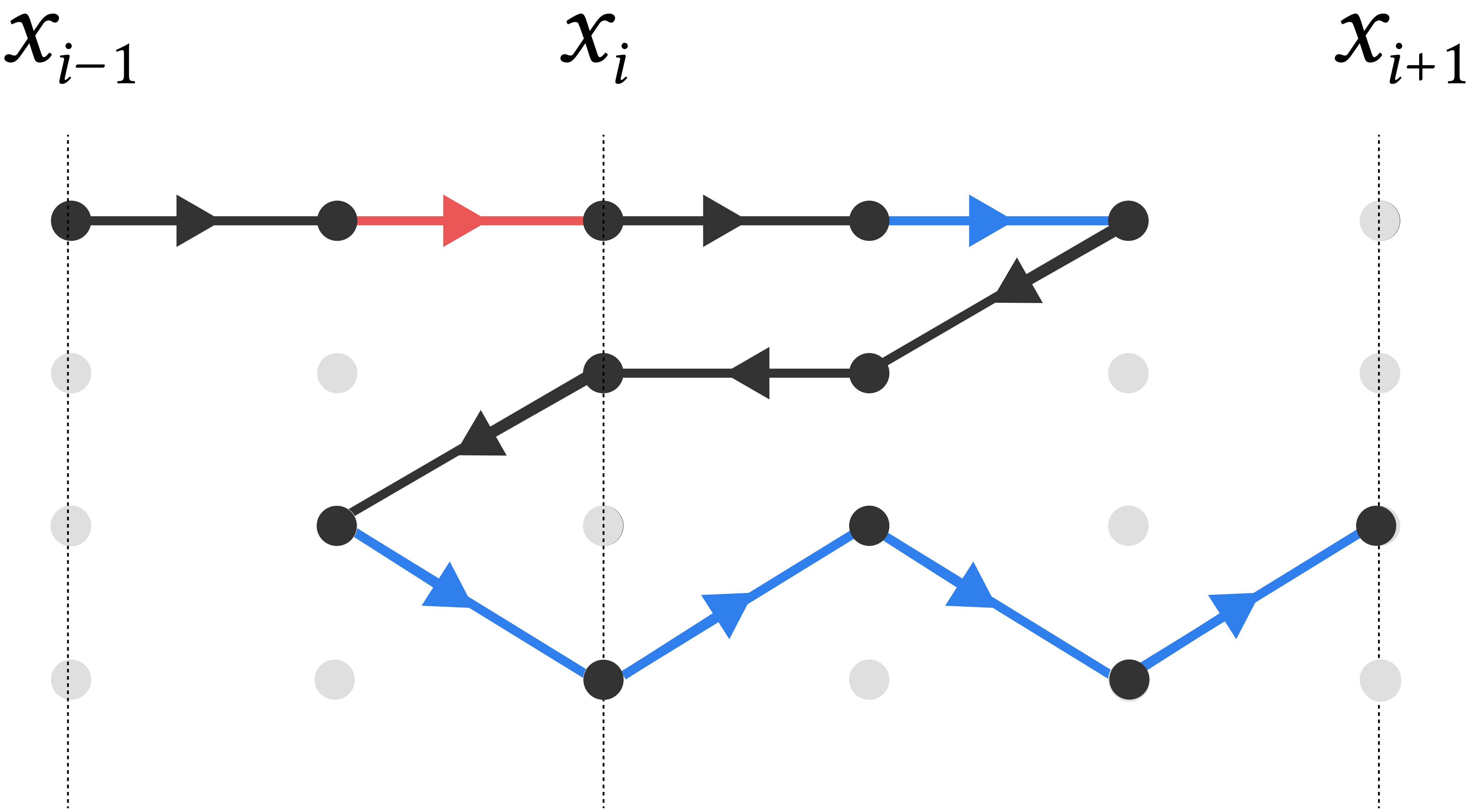
$$a \qquad \# \qquad a \qquad b \qquad b \qquad \# \qquad b \qquad a \qquad a$$

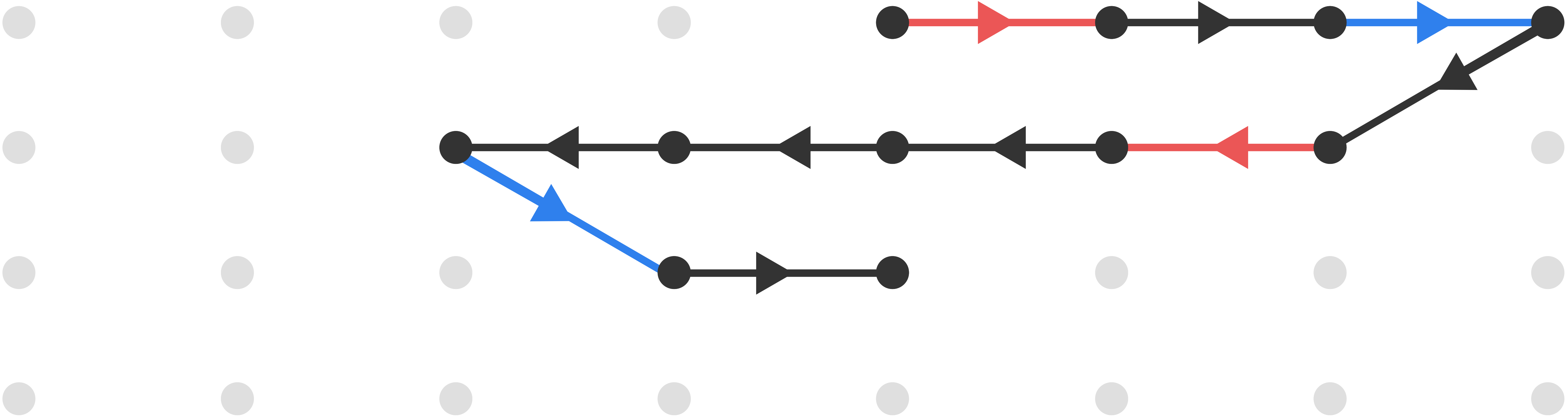
$$
\begin{tikzcd}
q_1 \arrow[r] \arrow[d, equal] & q_2 \arrow[ddr] & q_6 \arrow[r] \arrow[d, equal] & q_7 \arrow[r] & q_8 \arrow[r] & q_9 \arrow[ddr] & q_C \arrow[r] \arrow[d, equal] & q_D \arrow[r] & q_E \arrow[r] & q_F \\
q_1 \arrow[r, equal] & q_1 \arrow[d, equal] & q_6 \arrow[r, equal] & q_6 \arrow[r, equal] & q_6 \arrow[r, equal] & q_6 \arrow[d, equal] & q_C \arrow[r, equal] & q_C \arrow[r, equal] & q_C \arrow[r, equal] & q_C \arrow[d, equal] \\
q_0 \arrow[r] & q_1 & q_3 \arrow[r] & q_4 \arrow[r] & q_5 \arrow[r] & q_6 & q_9 \arrow[r] & q_A \arrow[r] & q_B \arrow[r] & q_C
\end{tikzcd}
$$

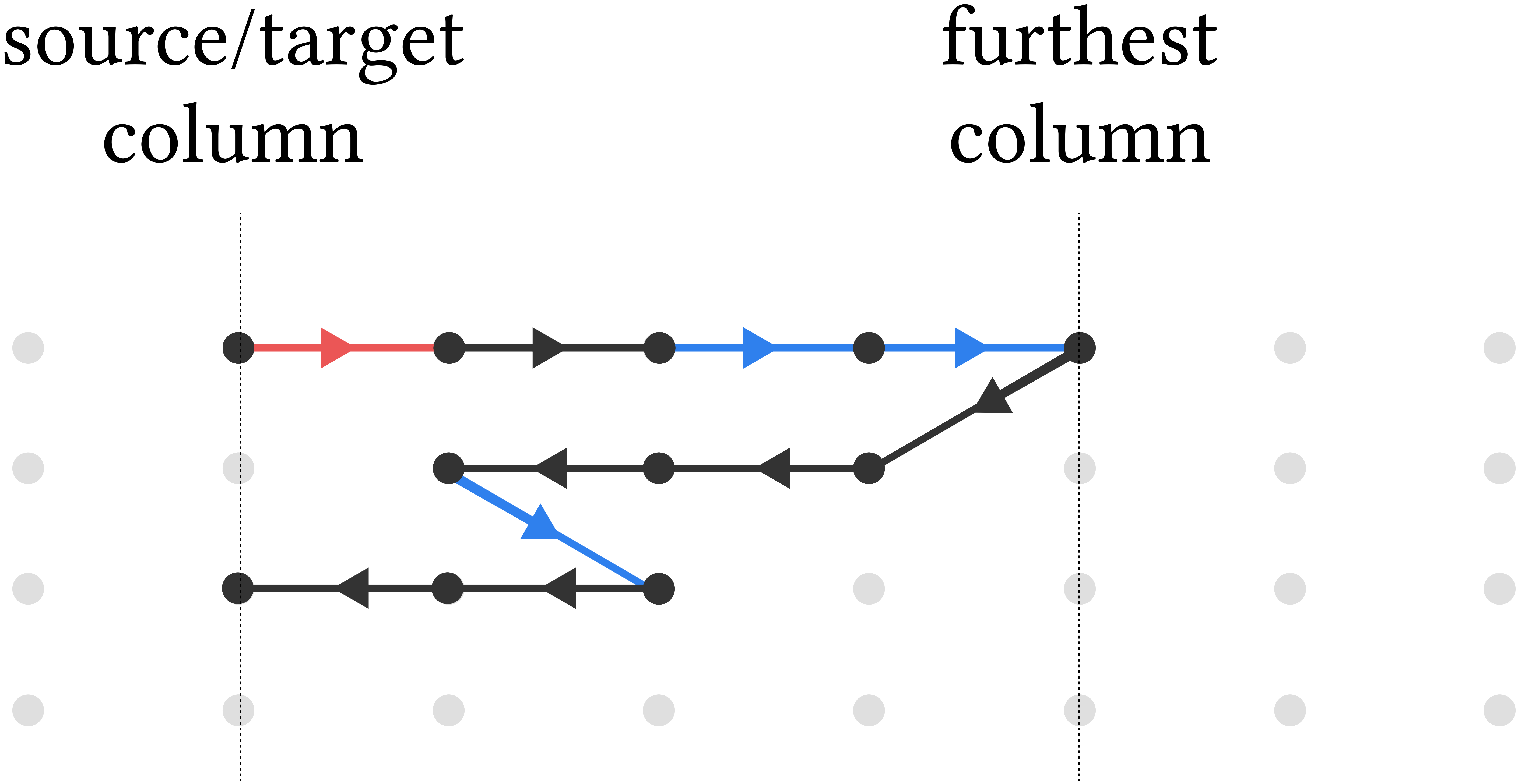

3
1
5
2
3
1
4
2
1
4
2
1
2
7

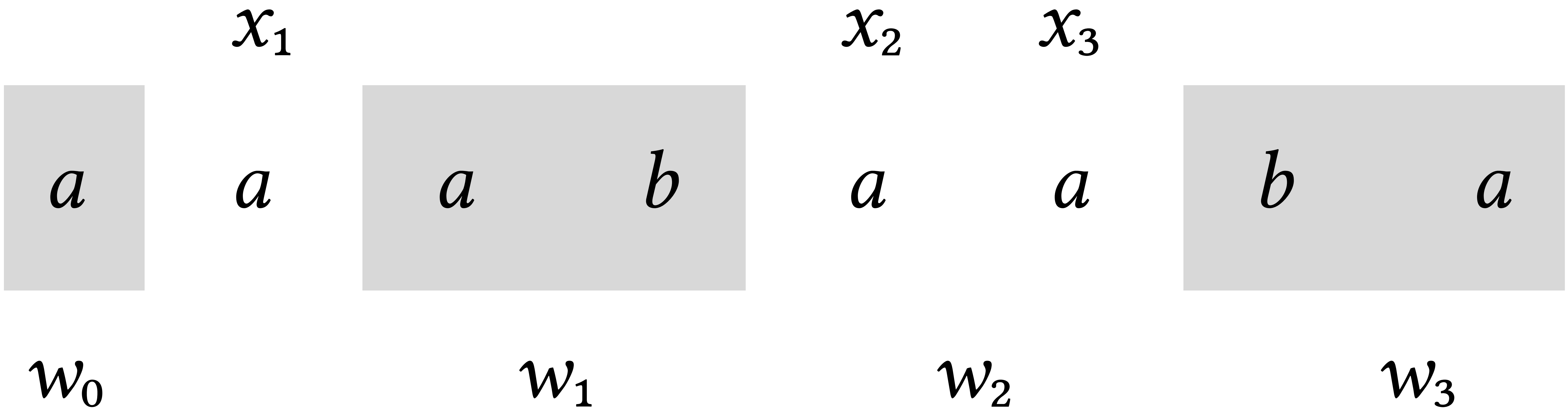


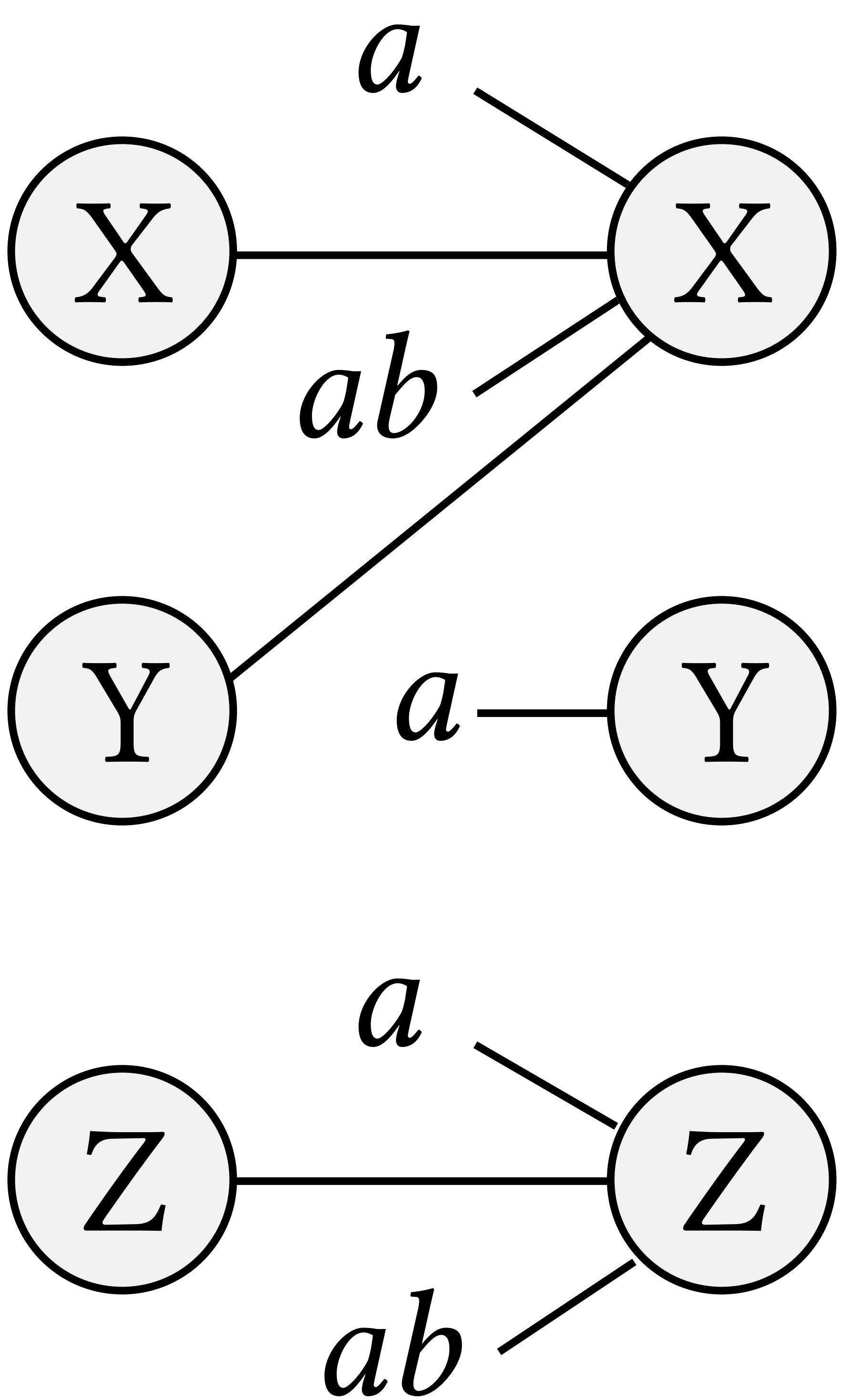

these columns are visited 3 times


x_{i-1}
x_i
x_{i+1}




source/target
column
furthest
column

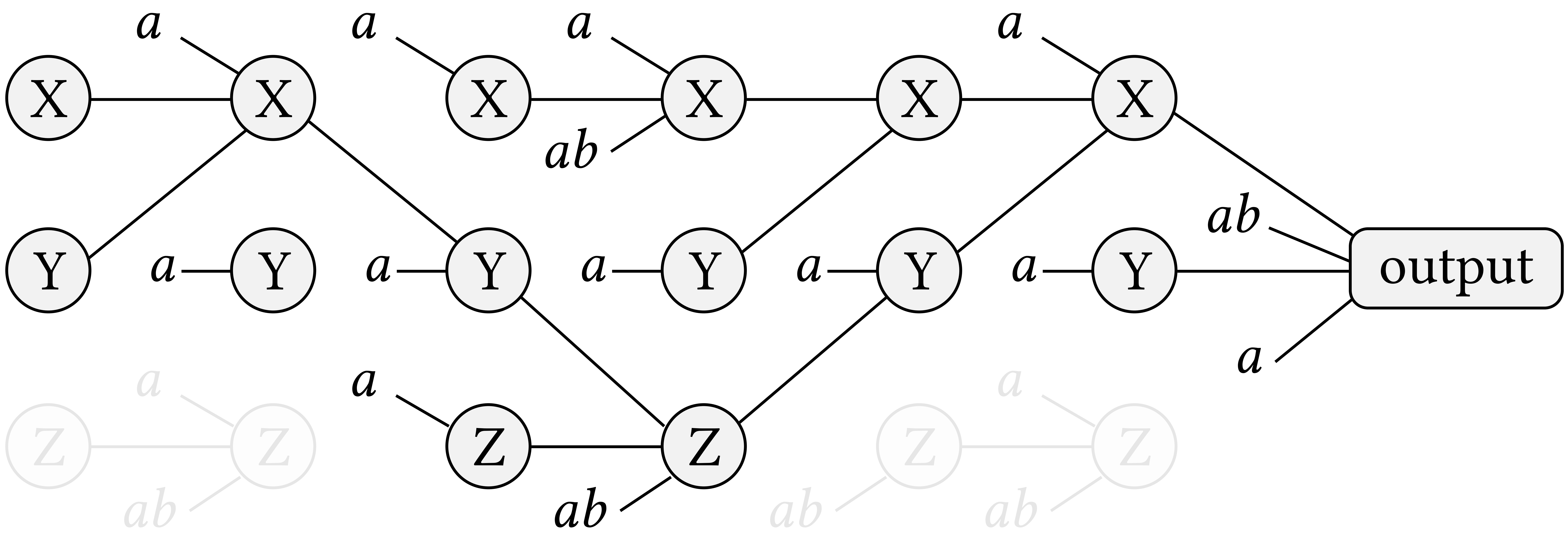

x1
x2
x3
a
a
a
b
a
a
b
a
w0
w1
w2
w3



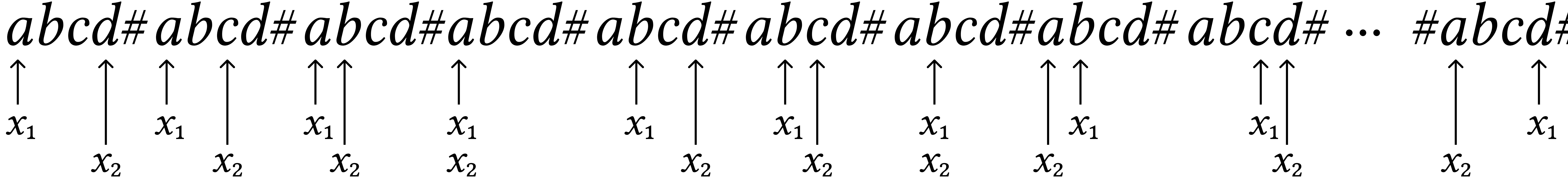
$$\mathrm{X} := a\mathrm{X}ab\mathrm{Y}$$

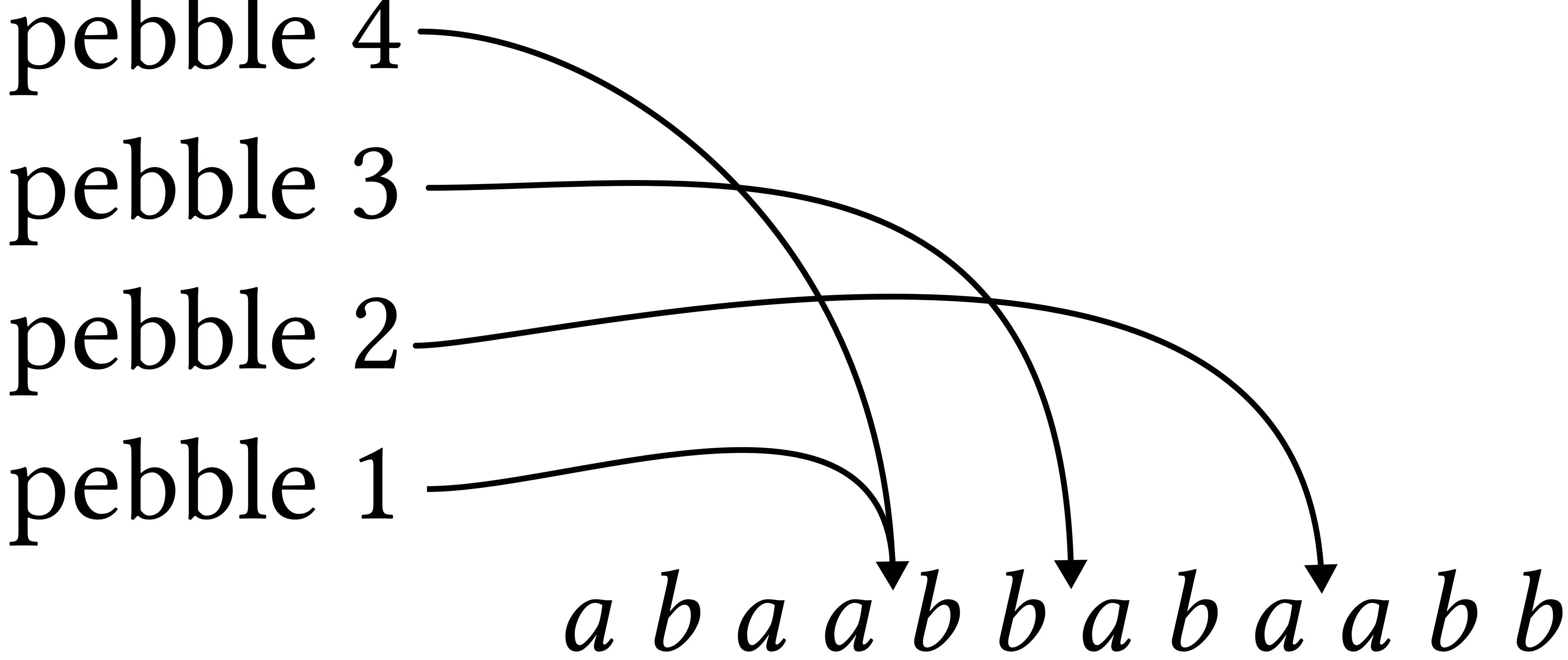
$$\mathrm{Y} := a$$

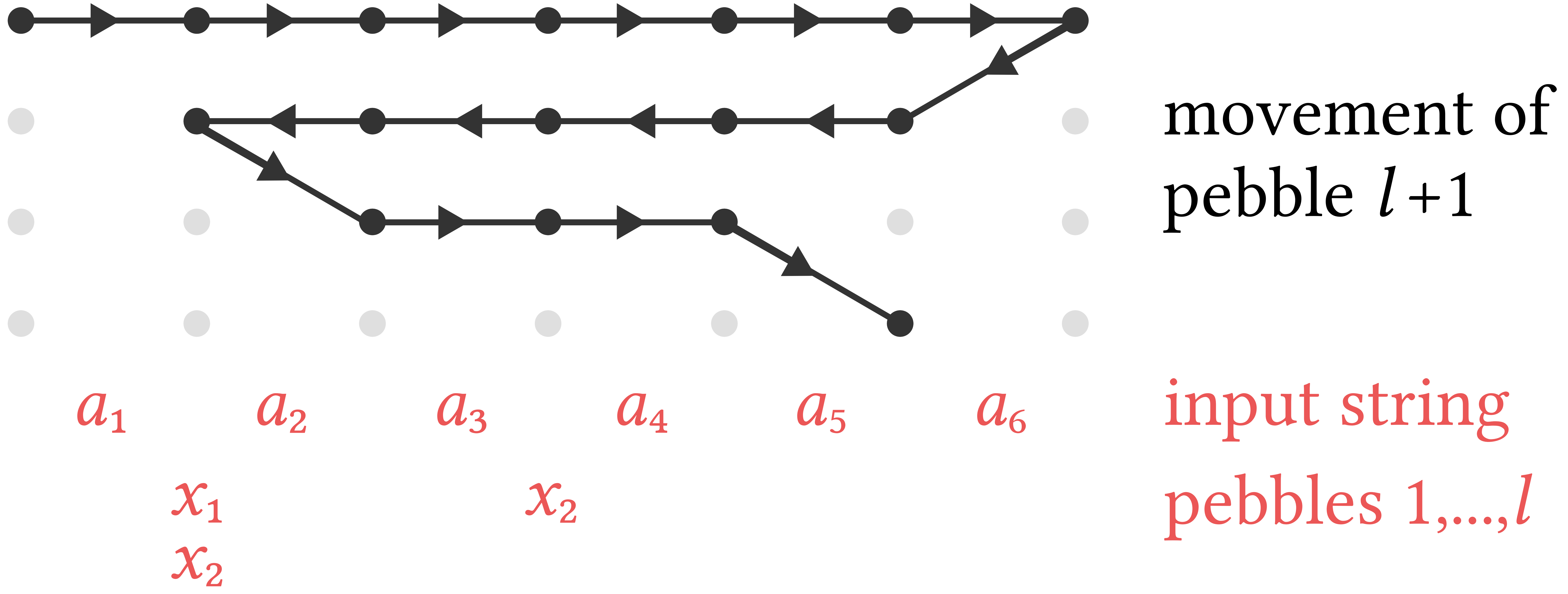
$$\mathrm{Z} := a\mathrm{Z}ab$$

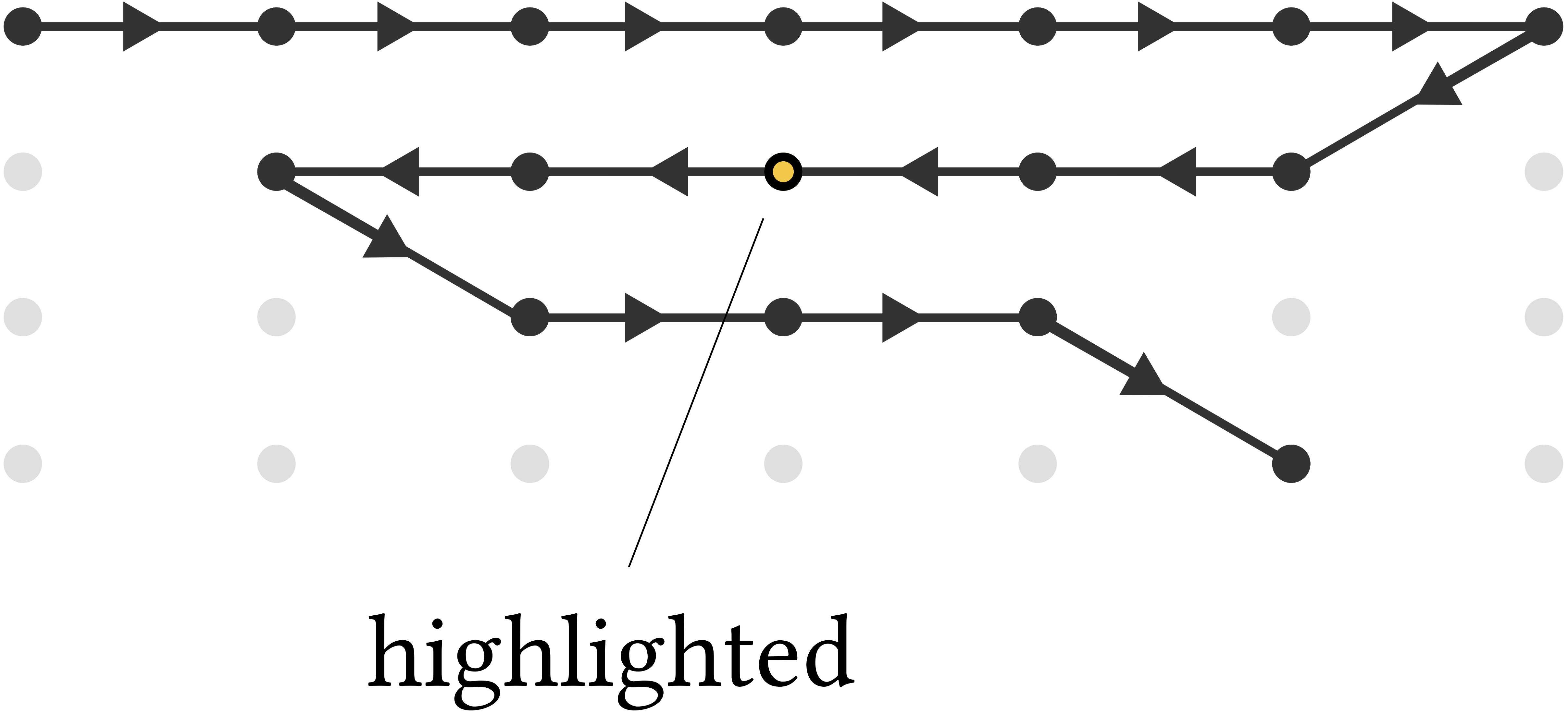


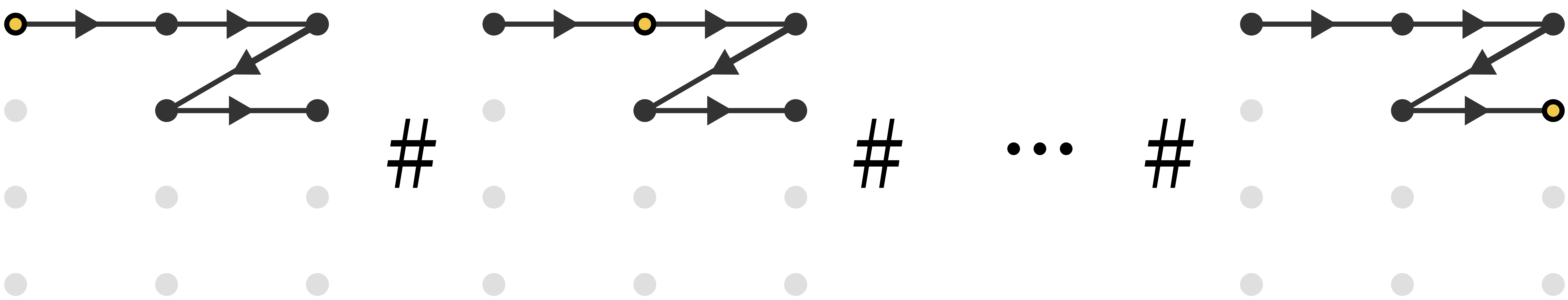
a
X
X
Y
a
Y
a
X
a
X
ab
a
Y
a
Z
Z
ab
a
Y
X
a
X
a
Y
a
Y
ab
output
a
a
Z
Z
ab
ab

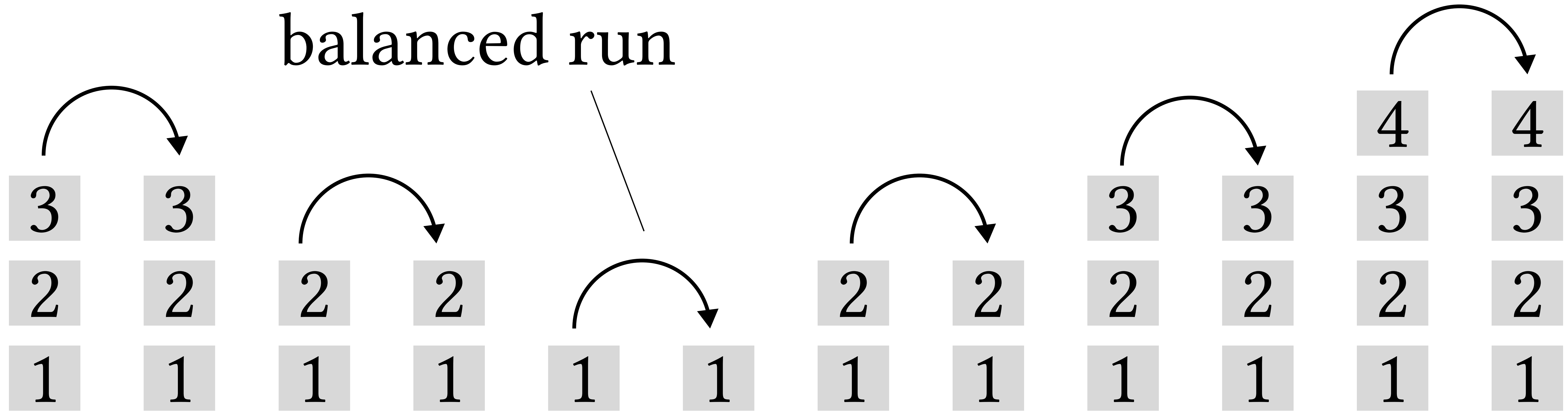

abcd# abcd# abcd#abcd# abcd# abcd# abcd#abcd# abcd# ··· #abcd#
x1 x2 x1 x2 x1 x2 x1 x2 x1 x2 x1 x2 x1 x2 x2 x1 x1 x2 x2 x1

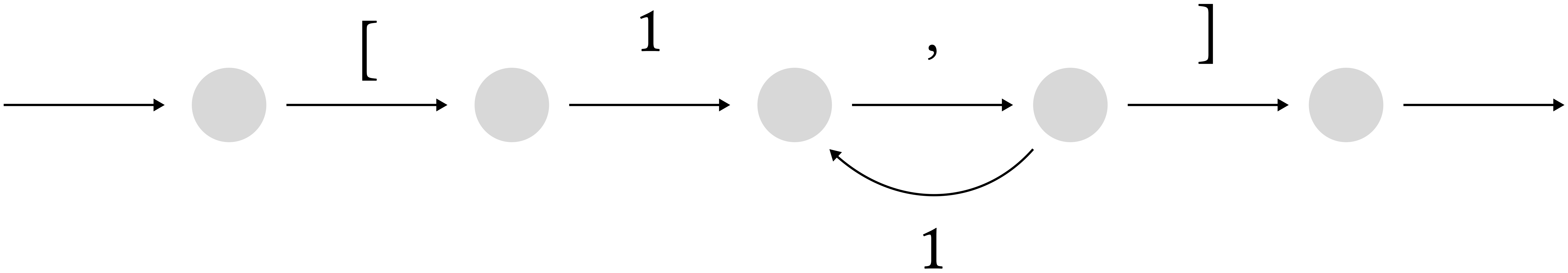

pebble 4
pebble 3
pebble 2
pebble 1
a b a a b b a b a a b b


movement of pebble $l+1$
a_1 a_2 a_3 a_4 a_5 a_6
input string
x_1 x_2
x_2
pebbles $1,\ldots,l$


highlighted




balanced run
3 3
2 2 2 2
1 1 1 1 1 1 1 1
2 2
3 3 4 4
2 2 3 3
1 1 2 2
1 1


[
1
,
]
1